Constellation

LaSalle County Station
2601 North 21st Road
Marseilles, IL 61341
815-415-2000 Telephone

RA24-007

10 CFR 50.73

April 26, 2024

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

LaSalle County Station, Unit 1
Renewed Facility Operating License No. NPF-11
NRC Docket No. 50-373

Subject: Licensee Event Report 2024-001-00, Core Operating Limits Report Thermal Limits Exceeded

In accordance with 10 CFR 50.73(a)(2)(i)(B), Constellation Energy Generation, LLC (CEG) is submitting Licensee Event Report (LER) Number 2024-001-00 for LaSalle County Station, Unit 1.

There are no regulatory commitments contained in this letter.

Should you have any questions concerning this report, please contact Ms. Laura Ekern, Regulatory Assurance Manager, at (815) 415-2800.

Respectfully,

Christopher J. Smith
Plant Manager
LaSalle County Station

Enclosure: Licensee Event Report 2024-001-00

cc: Regional Administrator – NRC Region III
NRC Senior Resident Inspector – LaSalle County Station

NRC FORM 366 (04-02-2024) U.S. NUCLEAR REGULATORY COMMISSION

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name		2. Docket Number	3. Page
LaSalle County Station, Unit 1	☒ 050 ☐ 052	00373	1 OF 3

4. Title
Core Operating Limits Report Thermal Limits Exceeded

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name		Docket Number
03	04	2024	2024	- 001 -	00	04	26	2024	NA	☐ 050	NA
									NA	☐ 052	NA

9. Operating Mode	10. Power Level
4	000 percent

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	**10 CFR Part 50**	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	**10 CFR Part 21**	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	**10 CFR Part 73**	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Joseph Reda, Operations Director	(815) 415-2200

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	AC	NA	G080	Y	NA	NA	NA	NA	NA

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On February 21, 2024, while performing the water rod inspection of fuel bundle GGJ319 during a planned refueling outage, it was identified that several fuel spacers had propagated up within the bundle toward the upper tie plate. It was determined through analysis that spacer positioning had a direct effect on the Minimum Critical Power Ratio (MCPR). On March 4, 2024, an analysis of the operating conditions by Global Nuclear Fuel (GNF) of fuel bundle GGJ319 was reviewed and accepted by the station. The report confirmed that during the previous cycle, the Operating Limit MCPR exceeded limits set by Technical Specifications (TS) 3.2.2. Since the Operating Limit MCPR was exceeded for longer than allowed by TS Limiting Condition for Operation (LCO) 3.2.2, this event is reportable as a condition prohibited by TS per 10 CFR 50.73(a)(2)(i)(B).

The fuel bundle spacers shifted out of position over the previous operating cycle. This condition was identified during the process of inspections in response to GNF 10 CFR Part 21 communication SC 22-04. This fuel bundle has been removed from the reactor core, is currently located in the spent fuel pool, and will not be re-loaded into the core.

There was no impact to the health and safety of the public or plant personnel because the Safety Limit MCPR was not exceeded in the previous cycle operation.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
LaSalle County Station, Unit 1	■ 050 / ☐ 052	00373	2024	- 001	- 00

NARRATIVE

Plant and System Identification

LaSalle Country Station Unit 1 is a General Electric Boiling Water Reactor with 3546 Megawatts Thermal Rated Core Power. The affected system was the reactor core system. The affected component was the GNF3 fuel bundle GGJ319.

Condition Prior to Event

Unit(s): 1	Date: March 4, 2024	Time: 00:00 CDT
Reactor Mode(s): 4	Mode(s) Name: Cold Shutdown	Power Level: 0 percent

Description

On February 21, 2024, during planned in-core fuel inspections for misoriented water rods of susceptible GNF3 fuel bundles in response to GNF 10 CFR Part 21 communication SC 22-04, an unexpected condition for fuel bundle GGJ319 was discovered where spacers 6, 7, and 8 were found near the top of the bundle upper tie plate. After initial discovery, subsequent inspection revealed that spacers 4 and 5 had also moved upward to the top of the fuel assembly. Fuel spacers 1 through 3 had not moved from their expected position. Fuel bundle GGJ319 was permanently discharged to the spent fuel pool.

Fuel vender GNF evaluated the impact of this condition for the previous operating cycle. The evaluation conservatively assumed the spacers were displaced for the entirety of the previous cycle, since it could not be determined when the spacers had moved. The evaluation was reviewed and accepted by the station on March 4, 2024.

The evaluation report determined that

- there were no penalties or adjustments associated with Linear Heat Generation Rate (LHGR) and Average Planar Linear Heat Generation Rate (APLHGR) and no discernable impact to thermal limits and margins.
- there was no discernable impact to thermal limits and margins for co-resident fuel in the Unit 1 reactor core during the previous cycle of operation.
- the Operating Limit MCPR exceeded limits set by TS 3.2.2 for the previous cycle.
- the Safety Limit MCPR requirements set by TS 2.1.1.2 were never violated for the previous cycle.

Cause

The apparent cause of the spacer relocation was determined to be the misoriented water rod in fuel bundle GGJ319. The misorientation of the water rod would have occurred during fuel bundle fabrication. The misorientation caused the spacers to not be locked into position, causing them to propagate up the bundle due to hydraulic force during normal operation.

Reportability and Safety Analysis

Fuel bundle GGJ319 must satisfy the requirements of TS 3.2.1, AVERAGE PLANAR HEAT GENERATION RATE (APLHGR); TS 3.2.2, MINIMUM CRITICAL POWER RATIO (MCPR); TS 3.2.3, LINEAR HEAT GENERATION RATE (LHGR); AND T.S. 2.1.1.2, SAFETY LIMITS. The report determined that the requirements of TS 3.2.1 for APLHGR and TS 3.2.3 for LHGR were met for the previous cycle operation. The report confirmed the requirements of TS 2.1.1.2 for the Safety Limit MCPR was never violated for the previous cycle operation.

1. FACILITY NAME	2. DOCKET NUMBER		3. LER NUMBER		
			YEAR	SEQUENTIAL NUMBER	REV NO.
LaSalle County Station, Unit 1	■ 050 / ☐ 052	00373	2024	- 001	- 00

NARRATIVE

The report determined that the requirements of TS 3.2.2 for the Operational Limit MCPR were not met for the previous cycle operation. LCO 3.2.2 requires all MCPRs to be greater than or equal to the MCPR operating limits specified in the Core Operating Limits Report. Condition A applies to any MCPR not within the limits and requires the MCPR(s) to be restored within the limits within 2 hours. In this case, the station had exceeded the completion time of 2 hours by assuming the condition existed since the beginning of the previous cycle. In addition, Condition B was not entered for the Completion Time of Condition A not being met. Therefore, this event is reportable in accordance with 10 CFR 50.73(a)(2)(i)(B) as a condition prohibited by the plant's TS.

Corrective Actions

Corrective actions taken in response to the conditions were:

- Sipped fuel bundle GGJ319 with both in-mast sipping and vacuum can sipping and it was identified that no leaking fuel rods were present.
- Performed full length peripheral visual inspection of dechanneled fuel bundle GGJ319.
- Evaluated the impact to LHGR, APLHGR, and MCPR limits.
- Permanently discharged fuel bundle GGJ319 to the spent fuel pool.
- Completed inspections on the remaining 271 fuel bundles in the Unit 1 susceptible population and confirmed no additional fuel bundles had a misoriented water rod.
- Confirmed no impact to Unit 2 due to no fuel bundles in the susceptible population.
- GNF implemented the following additional checks during the fuel fabrication process:
  - an in-process laser check on the orientation of the water rod tab,
  - removed the Quality at the Source process and reintroduced Quality Inspectors to the bundle assembly tables, and
  - added a check for tab orientation at final inspection.
- Constellation Energy Generation, LLC (and LaSalle County Station) enhanced the new fuel receipt inspection procedures to confirm proper water rod orientation on all new GNF3 fuel bundles.

Previous Occurrences

No previous occurrences.

Component Failure Data

Manufacturer: Global Nuclear Fuel (GNF)
Device: Unit 1 Fuel Bundle
Component ID: GGJ319